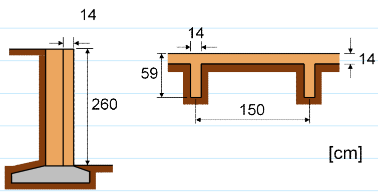
Muro de Arrimo com Contrafortes

- Dimensione o muro de arrimo
  - ⇒Blocos cerâmicos: $f_{bk} = 10\ MPa$
  - ⇒Solo: $K_a = 0{,}4$ e $\gamma = 20\ kN/m^3$



Esbeltez:

$h_e = 2 \cdot 260 = 520\ cm$

$e_{em} = 14\ cm$ $\quad t_{em}/t = 59/14 = 4{,}2$

$l_{em} = 150\ cm$ $\quad l_{em}/e_{em} = 150/14 = 10{,}7$

$t_{em} = 59\ cm$ $\quad \delta = \frac{1{,}2 - 1{,}4}{15 - 10}(10{,}7 - 10) + 1{,}4 = 1{,}37$

$t = 14\ cm$

$t_e = 1{,}37 \cdot 14 = 19{,}18\ cm$

$\lambda = \frac{h_e}{t_e} = \frac{520}{19{,}18} = 27{,}11 < 30$ aprovado p/ alvenaria armada

Empuxo

Topo: $\sigma_v = 0\ KN/m^2$ $\quad \sigma_{ha} = 0\ KN/m^2$

Base: $\sigma_v = 20 \cdot 2{,}60 = 52\ KN/m^2$ $\quad \sigma_{ha} = 0{,}4 \cdot 52 = 20{,}8\ KN/m^2$

Modelos estruturais com ações características:

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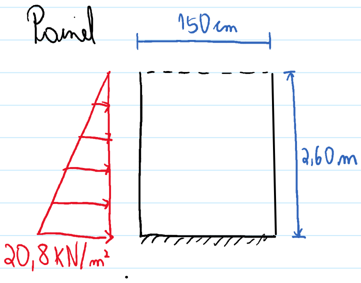
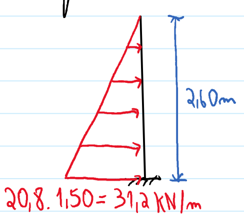
Painel

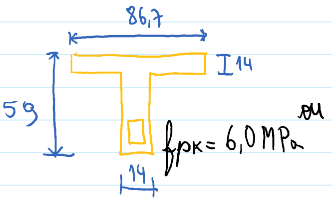


Contraforte



Projeto do enrijecedor

Geometria



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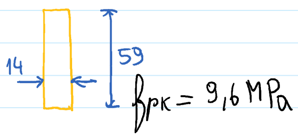


Solicitações

$V_K = 31{,}2 \cdot 2{,}60/2 = 40{,}56\ KN$

$M_K = 40{,}56 \cdot 2{,}60/3 = 35{,}15\ KNm$

Dimensionamento à flexão

$M_{sd} = 1{,}4 \cdot 35{,}15 = 49{,}21\ KNm = 4921\ KNcm$

$b_f = 86{,}7\ cm$

$h_f = 14\ cm$

$h = 59\ cm$

$d = 59 - 7 = 52\ cm$

$K f_d = 1{,}0 \cdot \frac{0{,}70 \cdot 0{,}6}{2{,}0} = 0{,}21\ KN/cm^2$

(1): $0{,}8x\, b\, Kf_d - f_{sd} \cdot A_s = 0$

(2): $0{,}8x\, b\, Kf_d \left(d - \frac{0{,}8x}{2}\right) = M_{sd}$

(2) $\longrightarrow x \longrightarrow$ (1) $\longrightarrow A_s$

(2): $0{,}8 \cdot x \cdot 86{,}7 \cdot 0{,}21\left(52 - \frac{0{,}8x}{2}\right) = 4921$

$x = 6{,}86\ cm \leq 0{,}45d = 0{,}45 \cdot 52 = 23{,}4\ cm$ OK!

$0{,}8x = 0{,}8 \cdot 6{,}86 = 5{,}49\ cm < h_f = 14\ cm$ OK!

$f_{sd} = 50/1{,}15 = 43{,}48\ KN/cm^2$ válida até Ø 10

(1): $0{,}8 \cdot 6{,}86 \cdot 86{,}7 \cdot 0{,}21 - 43{,}48 \cdot A_s = 0 \longrightarrow A_s = 2{,}30\ cm^2$ 3Ø10

$A_{s,mín} = \frac{0{,}15}{100} \cdot 14 \cdot 59 = 1{,}24\ cm^2$

Esforço cortante

$$\rho = \frac{A_s}{b\,d} = \frac{2,36}{14 \cdot 52} = 0,003241$$

$$f_{vk} = 0,35 + 17,5\,\rho = 0,35 + 17,5 \cdot 0,003241 = 0,407\ MPa$$

$$f_{vd} = \frac{f_{vk}}{\gamma_m} = \frac{0,407}{2,0} = 0,203\ MPa = 0,0203\ KN/cm^2$$

$$V_{sd} = 1,4\,V_k = 1,4 \cdot 40,56 = 56,78\ KN$$

$$\tau_{vd} = \frac{V_d}{b\,d} = \frac{56,78}{14 \cdot 52} = 0,078\ KN/cm^2 > f_{vd} = 0,0203\ KN/cm^2$$

precisa armar ao cisalhamento

$$V_a = f_{va}\,b\,d = 0,0203 \cdot 14 \cdot 52 = 14,78\ KN$$

$$f_{vd_2} = \frac{\sqrt{f_{pk}}}{\gamma_m} = \frac{\sqrt{9,6}}{2,0} = 1,55\ MPa = 0,155\ KN/cm^2$$

$$V_{Rd_2} = V_a + 0,4\,f_{vd_2}\,b\,d = 14,78 + 0,4 \cdot 0,155 \cdot 14 \cdot 52 = 59,92\ KN$$

$V_{sd} = 56,78\ KN < V_{Rd_2} = 59,92\ KN$ bielas comprimidas estão em boas condições

$$\frac{A_{sw}}{s} = \frac{V_{sd} - V_a}{0,75\ f_{yd} \cdot d} = \frac{56,78 - 14,78}{0,75 \cdot 43,48 \cdot 52} = 0,0248\ cm^2/cm = 2,48\ cm^2/m$$

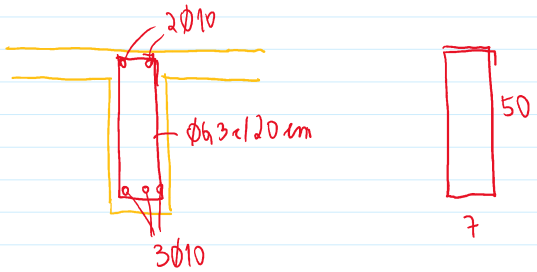
$$\left(\frac{A_{sw}}{s}\right)_{mín} = \rho_{mín}\,b = \frac{0,084}{100} \cdot 14 = 0,0118\ cm^2/cm = 1,18\ cm^2/m$$

Adotando estribo de dois ramos c/20 cm

$$\frac{A_{sw}}{s} = \frac{2 \cdot \pi \cdot \phi^2/4}{0,20} = 2,48\ cm^2/m$$

$$\phi = \sqrt{\frac{0,20 \cdot 2,48 \cdot 4}{2\pi}} = 0,56\ cm$$

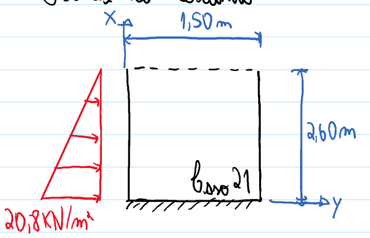
$\phi 6,3$ c/20 cm



Projeto do painel

Solicitações

Tabela de Bilônio



$l_a = 2,60\ m$

$l_b = 1,50\ m$

$\gamma = l_a/l_b = 2,60/1,50\ m = 1,73$

$p_d = 1,4 \cdot 20,8 = 29,12\ KN/m^2$

γ	μ_x	μ'_x	μ_y	μ_{yb}
1,70	1,65	7,85	3,98	2,80
1,75	1,64	7,95	4,09	2,78

$$\mu_x = \frac{1,64 - 1,65}{1,75 - 1,70}(1,73 - 1,70) + 1,65 = 1,644 \qquad M_{dx} = 1,644 \cdot 0,655 = 1,077\ KNm/m$$

$$\mu'_x = \frac{7,95 - 7,85}{1,75 - 1,70}(1,73 - 1,70) + 7,85 = 7,91 \qquad M_{dxe} = 7,91 \cdot 0,655 = 5,183\ KNm/m$$

$$\mu_y = \frac{4,09 - 3,98}{1,75 - 1,70}(1,73 - 1,70) + 3,98 = 4,046 \qquad M_{dy} = 4,046 \cdot 0,655 = 2,651\ KNm/m$$

$$\mu_{yb} = \frac{2,78 - 2,80}{1,75 - 1,70}(1,73 - 1,70) + 2,80 = 2,788 \qquad M_{ydB} = 2,788 \cdot 0,655 = 1,827\ KNm/m$$

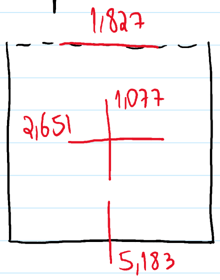
$1,75 - 1,70$

$M_{yb} = \frac{2,78 - 2,80}{1,75 - 1,70}(1,73 - 1,70) + 2,80 = 2,788$ $M_{ydB} = 2,788 \cdot 0,655 = 1,827 \ kNm/m$

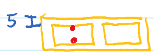
$l = \min(l_a; l_b) = 1,50 \ m$

$\frac{pl^2}{100} = \frac{29,12 \cdot 1,50^2}{100} = 0,655 \ kNm/m$

Momentos fletores de cálculo (kNm/m)



Dimensionamento à flexão



$d = 14 - 5 = 9 \ cm$

Adotando armaduras excêntricas simétricas

Flexão vertical

$M_{sd} = 518,3 \ kN \, cm/m$

$f_{pk,liq} = 0,6/0,48 = 1,25 \ kN/cm^2$

$K = 1,0$

$b_m = 15 \ cm$	$A_s = 0,22 \ cm^2$	Ø6,3 c/15 cm
$b_m = 30 \ cm$	$A_s = 0,43 \ cm^2$	Ø8 c/30 cm
$b_m = 45 \ cm$	$A_s = 0,65 \ cm^2$	Ø10 c/45 cm //
$b_m = 60 \ cm$	$A_s = 1,15 \ cm^2$	Ø12,5 c/60 cm

Flexão horizontal

$d = 7 \ cm$

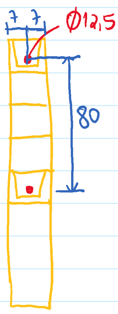
$M_{sd} = 265,1 \ kN \, cm/m$

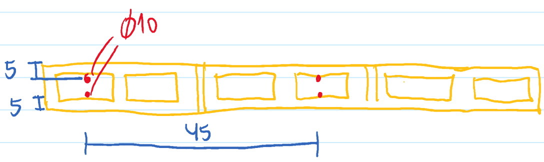
$f_{pk,liq} = 1,25 \ kN/cm^2$

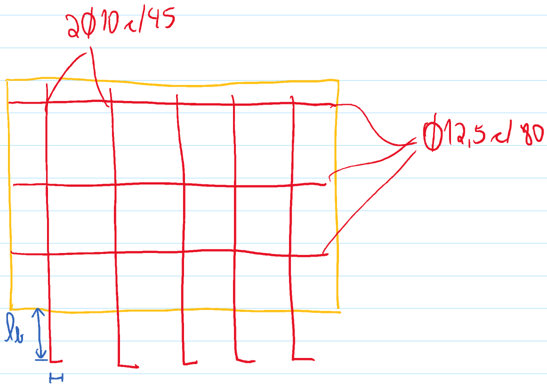
$K = 0,5$

$b_m = 20 \ cm$	$A_s = 0,21 \ cm^2$	Ø6,3 c/20 cm
$b_m = 40 \ cm$	$A_s = 0,42 \ cm^2$	Ø8 c/40 cm
$b_m = 80 \ cm$	$A_s = 1,03 \ cm^2$	Ø12,5 c/80 cm //

Adotando: vertical Ø10 c/45 cm
horizontal Ø12,5 c/80 cm







l_b
15